

Sarthak Pal

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SUMMARY

B.Tech CS student (2027) with two production-deployed applications and real users. Built a live AI workflow engine integrating Gemini API, DAG execution, and background job processing from scratch, solo, in under 3 weeks. Led a 4-person team at Smart India Hackathon 2025. AWS Certified AI Practitioner. Anthropic Claude certified. LeetCode 1650+, Top 15% globally. Comfortable shipping full-stack products, building AI backends, and debugging production failures under pressure.

EDUCATION

Prestige Institute of Engineering Management and Research (RGPV University) Indore, India
Bachelor of Technology in Computer Science Sep 2024 – Aug 2027
– Relevant Coursework: Data Structures & Algorithms, Systems Design, Large Scale Databases, Object-Oriented Programming

Shri Vaishnav Polytechnic College (RGPV University) Indore, India
Diploma in Computer Science – CGPA: 8.12/10.0 Sep 2021 – Jul 2024

TECHNICAL SKILLS

Frontend: React, Next.js 15 (App Router), TypeScript, Tailwind CSS, React Flow, Framer Motion
Backend: Node.js, Express, FastAPI, Python, REST APIs, JWT Auth, Async Processing, Docker
AI & APIs: Gemini API, LangChain, HuggingFace, Anthropic Claude, OpenAI API, Prompt Engineering
Databases & Tools: PostgreSQL, MongoDB, MySQL, Prisma, Git, GitHub Actions, Vercel, AWS (AI Practitioner)

PROJECTS

NextFlow – AI Workflow Engine | *Next.js 15, TypeScript, Trigger.dev, Gemini API, PostgreSQL* Apr 2026
– Built solo from idea to live product in under 3 weeks. Visual canvas for chaining LLM calls (Gemini Pro/Flash), image processing, and data transforms into DAG pipelines with real-time execution feedback
– Custom DAG engine: cycle detection, topological ordering, parallel background job dispatch via Trigger.dev. Clerk auth, Prisma schema, persistent run history in PostgreSQL. Live on Vercel
– Type-safe node I/O between pipeline steps enforced at compile time. Drag-and-drop node editor built with React Flow, dynamic edge routing, and live status indicators per node during execution

Sahayak – Agentic AI Platform | *Python, FastAPI, LangChain, HuggingFace, Docker* Sep 2025 – Oct 2025
– Agentic pipeline: NLP model classifies user intent, confidence-gated routing to 15+ specialized AI executors, fallback logic for low-confidence inputs. Led 4-person team at Smart India Hackathon 2025
– Diagnosed 15–20% timeout failures in production; traced root cause to synchronous LLM inference blocking the async event loop; rebuilt to queue-based architecture; failures dropped to <0.1%, uptime 99.2%
– Containerised with Docker for consistent deployment. Iterated on system prompts and confidence thresholds across multiple rounds to improve intent classification accuracy

Rebound – Full-Stack Social AI App | *React, Vite, Tailwind CSS, Node.js, Express, MongoDB* Feb 2026
– Complete product shipped in 48 hours at AMUHACKS 5.0: React frontend with Framer Motion animations, Node.js/Express REST backend, MongoDB, AI task prioritization engine, 3 role-based dashboards
– AI engine scores tasks by deadline urgency and stress level, serving personalized priority queues per user role. Top 8 out of 100+ teams. Live with real users on Vercel

Protec-X – Real-Time Detection System | *Java, MySQL, JDBC, Multithreading, MVC* Feb 2025 – Present
– Real-time anomaly detection in Java: background threads continuously monitor MySQL for suspicious patterns without blocking the main request thread. Heuristic detection logic built from scratch
– MVC architecture, role-based dashboards for admins and users, end-to-end schema design and deployment

ACHIEVEMENTS

Hackathons: AMUHACKS 5.0 – Top 8/100+ teams (Feb 2026) | Salesforce Dreamquest – Top 150 globally (2026)
Competitive Programming: LeetCode 400+ problems | Rating 1650+ | Top 15% globally
HackerRank: 5-Star Java | SQL Advanced certified
Certifications: AWS Certified AI Practitioner (Apr 2026) | Anthropic Claude Code in Action (2025)
Challenges: GfG 160 Days Challenge (2024) | Smart India Hackathon 2025 – Team Lead